SECURITY OPERATIONS CENTER (SOC) REPORT

Alert Management, Incident Response, and Threat Analysis

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Introduction

This report demonstrates Security Operations Center (SOC) activities including alert prioritization, incident classification, alert triage, SIEM monitoring, log analysis, and incident response. The objective is to simulate a brute-force SSH attack and analyze it using Wazuh dashboards and system logs.

These activities reflect real-world SOC operations where analysts detect, investigate, and respond to security incidents using logs, alerts, and monitoring tools.

1. Alert Priority Levels

Alert prioritization is essential in SOC operations to identify critical threats. Alerts are classified based on severity such as Critical, High, Medium, and Low depending on impact and exploitability.

The Common Vulnerability Scoring System (CVSS) is used to measure severity. High scores indicate serious risks that require immediate action.

Figure 1: Alert priority classification table

Alert ID	Alert Type	Asset	CVSS Score	Priority	Reason
1	Ransomware Detected	Prod Server	9.8	Critical	Data encryption risk
2	Unauthorized Admin Login	Web Server	8.5	High	Privilege abuse
3	Brute Force Attempt	SSH Server	6.5	Medium	Repeated login attempts
4	Port Scan	Test VM	3.2	Low	Recon activity

2. Incident Classification

Security incidents are categorized based on their type such as phishing, malware, or brute-force attacks.

In this scenario, the activity is identified as a brute-force attack under MITRE ATT&CK technique T1110.

Figure 2: Incident classification with MITRE mapping

Alert ID	Incident Type	Description	MITRE Technique	Severity
1	Phishing	Suspicious email with link	T1566	High
2	Malware	Ransomware detected	T1486	Critical
3	Brute Force	Multiple failed SSH logins	T1110	Medium
4	Port Scan	Multiple ports scanned	T1046	Low

3. Alert Triage

Alert triage is the process of analyzing alerts to determine whether they represent real threats or false positives.

Repeated failed login attempts from the same source confirmed a brute-force attack pattern.

Figure 3: Alert triage table

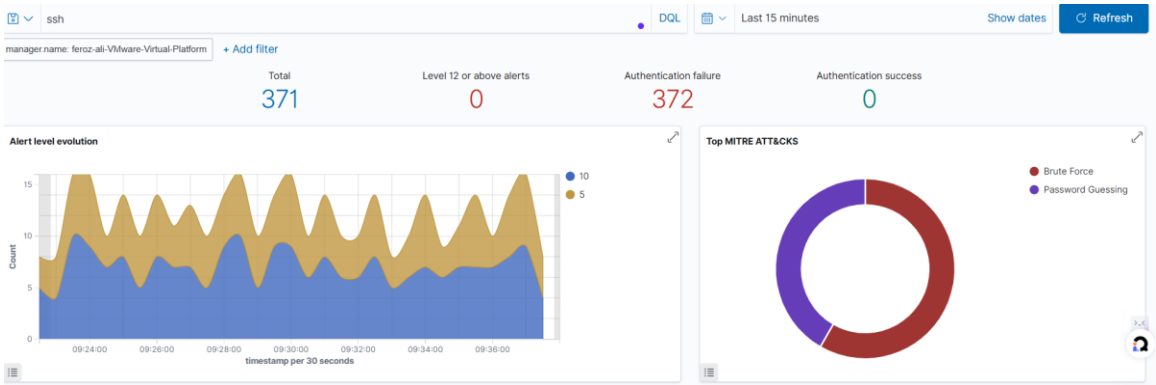
Alert ID	Type	Source IP	Target	Priority	MITRE Technique	Status
001	SSH Brute-force Attack	192.168.110.133	Ubuntu Server	High	T1110 (Brute Force)	Open

4. Wazuh Dashboard Analysis

Wazuh SIEM provides centralized monitoring and visualization of security events.

The dashboard showed a high number of authentication failures indicating suspicious activity.

Figure 4: Wazuh dashboard showing alerts

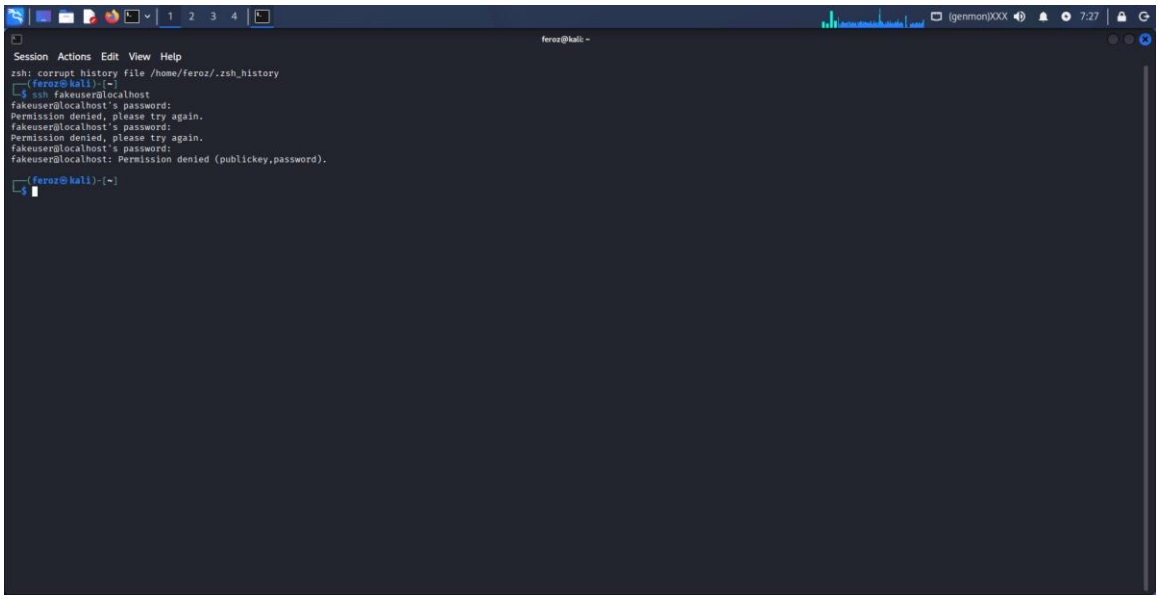


5. Incident Simulation

A brute-force SSH attack was simulated by performing multiple failed login attempts.

This helps in understanding how real attacks generate logs and alerts.

Figure 5: SSH attack simulation

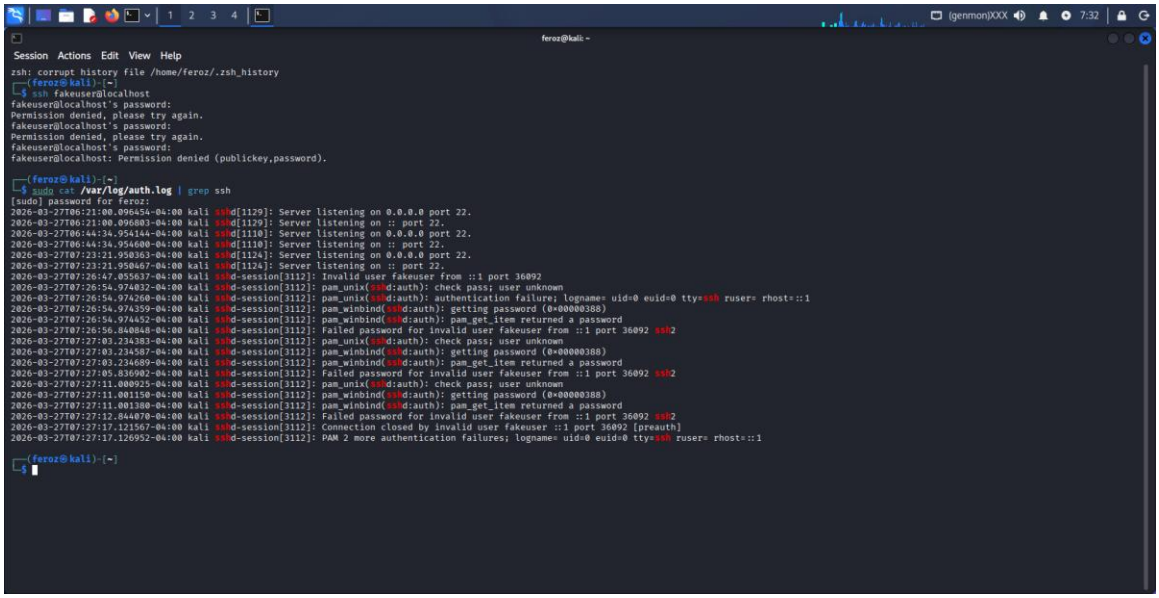


6. Log Analysis

Log analysis was performed using system authentication logs stored in /var/log/auth.log.

Multiple failed login attempts confirmed malicious activity.

Figure 6: Log analysis output

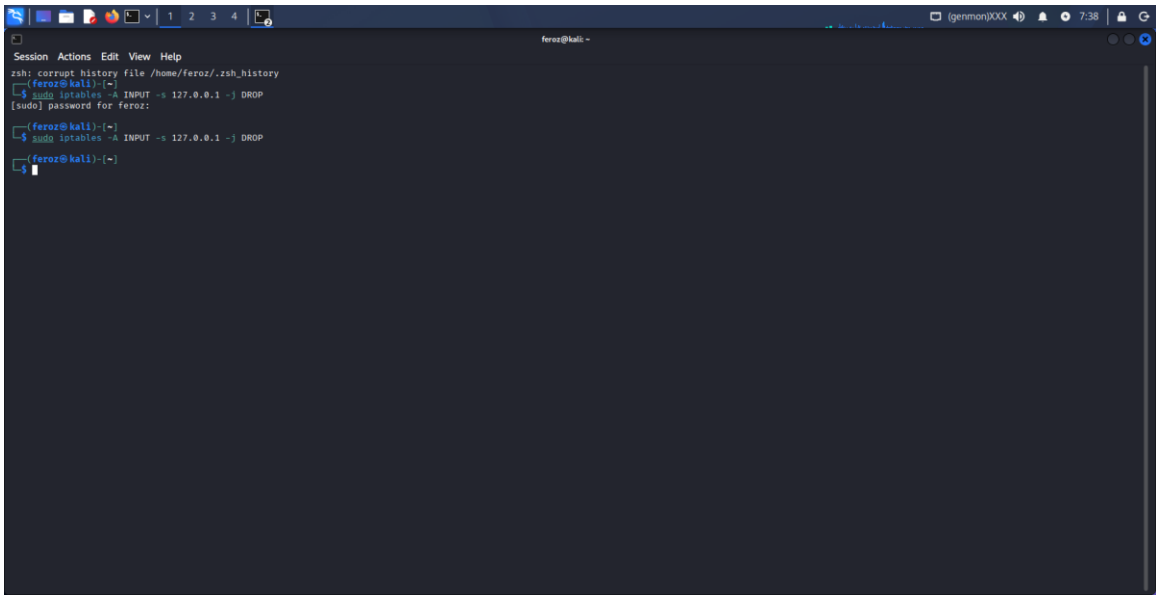


7. Incident Response

Incident response actions were taken to mitigate the attack.

Firewall rules were applied using iptables to block suspicious traffic.

Figure 7: Firewall rule applied



8. Evidence Collection

Evidence was collected from logs and verified using SHA256 hashing to ensure integrity.

This ensures that the collected data remains unchanged and valid for analysis.

Figure 8: Evidence collection table

Item	Description	Source	Collected By	Date	Hash Value
Auth Log	SSH brute-force attempts	/var/log/auth.log	Feroz Ali	2026-03-28	c353831b667f8984a856524f6829db71d33b183690af4440c9c85c59c2c4071b
Wazuh Alert	Authentication failure alerts	Wazuh Dashboard	Feroz Ali	2026-03-28	N/A
Terminal	Attack simulation output	Kali Terminal	Feroz Ali	2026-03-28	N/A

9. Incident Response Lifecycle

The incident followed a structured lifecycle including preparation, identification, containment, eradication, recovery, and lessons learned.

This ensures systematic handling of security incidents.

Figure 9: Incident response lifecycle

Key Learnings

1. Alert prioritization helps focus on critical threats.
2. Log analysis is essential for detecting brute-force attacks.
3. SIEM tools improve visibility of security events.
4. Incident response reduces impact of attacks.

References

Wazuh Documentation – https://documentation.wazuh.com
MITRE ATT&CK – https://attack.mitre.org
Linux Manual – https://man7.org/linux/man-pages/